

Vol 2 Chapter 10 — Neutrinos and the Seesaw Mechanism

Keeper (author pass)

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Chapter 10 — Neutrinos and the Seesaw Mechanism

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

Neutrinos are the lightest known massive particles in nature. They come in three flavors (electron, muon, tau), they oscillate between flavors as they propagate, and their masses — though now known to be non-zero from oscillation measurements — remain millions of times smaller than the masses of the charged leptons they pair with. The standard explanation for this enormous mass hierarchy is the seesaw mechanism: a heavy right-handed Majorana mass scale produces, via mixing, a light effective left-handed neutrino mass.

BST anchors the seesaw at BST primary integer 17 (substrate's natural seesaw scale) with the substrate-derivation tying neutrino masses to substrate K-type weights.

10.1 Neutrinos as substrate windings

Casey's W-55 work (May 2026) identifies neutrinos as the substrate's $SO(2)$ -trivial windings — Shilov-boundary primitive cycles that carry zero $SO(2)$ phase weight, hence zero electric charge. This is the substrate-side reading of why neutrinos are electrically neutral and why they couple only via the weak interaction.

The substrate's $SO(2)$ representation theory permits both charged windings (charged leptons, Chapter 5) and trivial windings (neutrinos). Three generations produce three neutrino species, one per generation pairing.

10.2 The seesaw at BST primary 17

The seesaw mass relation in standard physics:

$$m_\nu \sim \frac{m_D^2}{M_R},$$

with m_D a Dirac mass scale and M_R a heavy Majorana scale. The neutrino mass m_ν is suppressed by the ratio m_D/M_R .

BST identifies M_R with substrate primary integer 17 (a BST-derived integer entering at the seesaw scale: $17 = N_C^3 - \text{rank} \cdot n_C = 27 - 10$). The Dirac scale m_D is set by the charged-lepton mass scale; the seesaw suppression gives substrate-derived m_ν values at the appropriate millielectronvolt range.

The seesaw scale BST primary 17 is part of the framework's integer-web structure (Volume 0 Chapter 6). Its appearance in the neutrino sector is one of the integer web's substantive cross-domain anchorings.

10.3 PMNS mixing — the Triangle

The Pontecorvo–Maki–Nakagawa–Sakata (PMNS) matrix encodes neutrino flavor mixing. Three mixing angles and one CP-violating phase parameterize it; experimentally, the angles are large (compared to CKM quark mixing).

Lyra T1932 (the PMNS Triangle theorem) closes the substrate-derivation: the PMNS mixing angles are the substrate's three rank-2 K-type rotation angles between the three neutrino species, with the substrate-mechanism reading via Q^5 Chern-class flux on the neutrino-K-type sector.

The PMNS Triangle is the substrate's analog of the CKM unitarity triangle (Chapter 7). The two triangles share a common substrate-side origin in the rank-2 doublet structure that produces flavor mixing in both the quark and lepton sectors.

10.4 Neutrino masses and the mass hierarchy

The substrate-derived neutrino masses span:

- $m_{\nu_1} \sim 10^{-2}$ eV (lightest)
- m_{ν_2}, m_{ν_3} : at the seesaw-suppressed millielectronvolt range with the substrate Casimir-weighted hierarchy

Match precisions to current oscillation measurements are at the percent level. The neutrino mass hierarchy (normal vs inverted ordering, currently unresolved experimentally) is predicted by BST to be normal ordering — multi-year experimental falsifier.

10.5 Sterile neutrinos: predicted absence

A defining BST prediction is that no sterile neutrinos exist (Five-Absence Set, Chapter 11). The substrate's K-type structure exhausts the neutrino sector with the three

left-handed neutrinos paired with the charged leptons; no right-handed neutrino K-type exists in the substrate’s spectrum.

Recent experimental searches for sterile neutrinos (LSND anomaly, MiniBooNE follow-up, IceCube sterile-neutrino search) have produced ambiguous results. BST predicts they will not find what they are looking for; current bounds are consistent with this prediction.

10.6 Neutrinoless double beta decay and $0\nu\beta\beta$

BST predicts Majorana neutrinos: the mass is the $\Delta L = 2$ Weinberg operator on the chargeless S^4 locus, the right-handed neutrino is a total gauge singlet, and the odd substrate dimension $g = 7$ forbids a light ν_R Dirac partner. Neutrinoless double beta decay therefore occurs. With normal ordering, $m_1 = 0$, and the derived mixing angles $\sin^2 \theta_{12} = 3/10$, $\sin^2 \theta_{13} = 1/45$, the two Majorana phases sweep the effective mass over

$$|m_{\beta\beta}| \in [1.4, 3.7] \text{ meV}.$$

A detection in this window supports BST; a firm null below ~ 1 meV falsifies it. This is a next-generation target ($m_1 = 0$ suppresses $|m_{\beta\beta}|$ below the inverted-hierarchy scale).

10.7 What comes next

Chapter 11 — Beyond Standard Model: The Five Absences — develops the substrate’s five-absence prediction set (no GUT, no proton decay, no monopoles, no sterile neutrinos, no SUSY) as the framework’s headline falsifier package.

Where to look this up: Neutrinos as SO(2)-trivial windings: Casey W-55. PMNS Triangle: Lyra T1932. Seesaw at BST primary 17: substrate integer-web work. Sterile-neutrino prediction: Five-Absence Set. For standard neutrino-sector treatments: Mohapatra and Pal, *Massive Neutrinos in Physics and Astrophysics*; particle data group neutrino reviews.